

Detecting Developer Burnout in AI Startups With Remote Workforce: Analyzing Time Zones, Cultural Factors, and Work Patterns in Open-Source Repositories

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I. INTRODUCTION

Burnout—described as the emotional fallout of relentless, unmanaged workplace stress—is no longer a distant or abstract concern in software engineering. It’s a real, present, and growing threat that affects not just productivity, but the people behind the code. Marked by emotional exhaustion, growing detachment, and a deepening sense of ineffectiveness [3], [5], burnout rarely announces itself loudly. Instead, it creeps in quietly—eroding motivation, straining relationships, and dulling the creative spark that drives great software. Prior work has shown that burnout contributes to absenteeism, presenteeism, and decreased engagement in development teams [7], yet it remains underexamined in early-stage, high-pressure software environments. By the time its impact becomes visible in team performance or turnover, the damage is often already done.

This concern becomes even more urgent in the context of early-stage AI startups that are both remote-first and open-source-driven. These organizations sit at the bleeding edge of technological innovation, yet they often function without the structural safety nets found in more established firms. Developers in early-stage AI startups are often asked to do it all—manage themselves, ship code fast, and stay ahead of ever-changing tools and frameworks. They do this while juggling the uncertainty of remote teamwork and the constant visibility that comes with building in the open. Every line of code, every delay, every decision plays out not just within their team, but in front of a watching global community. In this environment, pressure doesn’t just build—it compounds. What starts as productive momentum can quietly tip into unsustainable overwork, with little room to pause or ask for help. As these startups race to define the digital future, understanding how burnout takes hold in their teams isn’t just useful—it’s essential to creating more humane, sustainable ways of working.

Early-stage AI startups represent an ideal yet understudied

setting for examining developer burnout. These companies are uniquely positioned at the intersection of innovation, speed, and uncertainty. Unlike more mature firms, they often operate without dedicated HR or management infrastructure—placing the burden of coordination, communication, and output squarely on the developers themselves. The AI domain compounds these pressures, as developers must continually learn and adapt to rapidly evolving tools, models, and frameworks just to stay competitive. This cognitive load is distinct from traditional software work, requiring continuous experimentation with complex systems under high stakes. Open-source AI startups, in particular, face the additional challenge of building in public, where performance is constantly scrutinized by an external developer community. Despite these unique pressures, little research has explored burnout in the context of early-stage AI development. This confluence of technical complexity, public visibility, and limited support makes burnout not just a possibility—but a structural risk that deserves empirical attention.

This study focuses on a specific and understudied group: developers working remotely in newly funded AI startups that maintain public open-source repositories. These settings are particularly important to investigate for several reasons. First, AI startups often rely on small, distributed teams that must self-manage while delivering high output under tight timelines. Second, the open-source model adds public visibility and community expectations to already high internal pressures. Developers in such contexts frequently deal with unclear role boundaries, schedule unpredictability, and high expectations around availability and output—all of which have been linked to burnout [21]. Despite the relevance of these factors, little empirical work has explored burnout in this particular context using behavioral data.

The goal of our study is to identify early indicators of burnout through developers’ work habits, using publicly available data from version control systems. While much of the

existing research relies on surveys or interviews, our approach is grounded in objective behavioral signals: GitHub commit activity, sentiment analysis, and time-series-based anomaly detection. This novel combination allows us to track changes in developer behavior—such as reduced activity, emotional tone in commit messages, or inconsistent working hours—that might point to the onset of burnout.

Our work builds on prior studies that examined developer productivity [19], sentiment in commit messages [20], and the structure of open-source contributions [22]. However, few studies have integrated these dimensions to construct a unified view of burnout risk. We address this gap by focusing specifically on early-stage AI startups—organizations operating at the cutting edge of technology, yet highly vulnerable to team instability due to burnout. Understanding how burnout manifests in these settings is not only academically relevant but practically urgent, given their role in driving innovation across industries.

By analyzing these cases, we aim to contribute to the growing field of empirical software engineering by offering a scalable, reproducible method to monitor developer well-being. We envision future applications in automated tooling that supports team leads and project maintainers in detecting burnout before it disrupts productivity or morale. Ultimately, this research supports broader efforts to design healthier and more sustainable work environments in a remote-first, innovation-driven software industry.

II. RESEARCH QUESTIONS

This study is guided by the following research questions, designed to investigate how behavioral signals captured from GitHub activity relate to developer burnout. Each question is motivated by gaps in the current literature and theories about stress, workload, and organizational behavior.

RQ1: Can commit activity and sentiment analysis predict periods of developer burnout in startup repositories?

Most of what we know about burnout comes from self-reported surveys and retrospective interviews—methods that rely on people recognizing and naming their own struggles, often after the fact. But burnout doesn’t always show up in ways people can easily articulate. It builds quietly, reflected in small shifts: working late into the night, slowing down, sounding less upbeat in a commit message. This question asks whether we can catch those signs earlier—by listening not just to what developers say, but to how they work. By analyzing behavioral traces like commit patterns and emotional tone, we explore whether burnout can be predicted before it fully takes hold. Our approach builds on occupational stress theory and prior empirical work [19], [20], extending it into a longitudinal, data-driven model that can potentially detect burnout before it becomes disruptive.

This question is motivated by the specific gap in the current literature we examined: even though prior work has used sentiment analysis to study software team morale [19], [20], few studies have combined temporal commit activity with

sentiment signals to track longitudinal indicators of burnout. Moreover, existing methods typically examine individual metrics in isolation. Our approach integrates these signals to assess whether subtle shifts in behavior—before any self-report—can signal the onset of burnout, contributing to the development of more scalable monitoring tools for well-being.

RQ2: How does a developer’s level of responsibility within an open-source AI startup (e.g., founder, core maintainer, occasional contributor) shape their vulnerability to burnout, as reflected in behavioral signals?

People that are in high-responsibility roles often face overlapping demands—such as writing code and managing the team at the same time. In early-stage AI startups, these demands are intensified by rapid iteration cycles and limited support structures. Organizational behavior theory [3] suggests that role ambiguity and responsibility overload are key predictors of burnout, yet few studies have examined how these stressors manifest behaviorally in open-source contexts. By comparing commit behaviors and sentiment across contributor roles, we test whether leadership comes at a psychological cost—even in communities that prize autonomy and openness.

While organizational psychology has long highlighted role overload and ambiguity as burnout predictors [3], there is limited empirical evidence on how these dynamics play out in open-source software development. Previous research often treats developers as a homogenous group, overlooking how different levels of responsibility may shape work patterns and stress exposure. By analyzing behavioral signals across contributor roles, we respond to this theoretical gap and offer a more nuanced understanding of how leadership and ownership may come at a psychological cost—even in decentralized, volunteer-driven communities.

RQ3: How do irregular contribution rhythms—such as late-night or weekend work—correlate with behavioral signals of burnout in open-source AI startups with globally distributed teams?

In remote-first AI startups, developers often set their own schedules. That freedom can be empowering, but over time, it can also lead to unhealthy work patterns—working into the night, pushing through weekends, or cycling between intense activity and sudden silence. These rhythms don’t always look like overwork at first. But when work stretches across all hours and recovery time disappears, something deeper may be going on. This question asks whether those patterns—when developers are working, not just how often—can help us detect early signs of burnout before they surface in more obvious ways.

Existing studies often focus on overall workload volume, but fewer investigate the structure and timing of work as potential burnout indicators. Behavioral science suggests that irregular routines and insufficient recovery periods can undermine mental health [12], yet this has not been thoroughly tested in the context of distributed software teams. Our question addresses this by examining whether inconsistent contribution

rhythms—such as weekend bursts or late-night coding—serve as early signals of developer strain. Unlike general discussions of remote work, our focus is on specific behavioral routines observable in open-source repositories.

III. LITERATURE REVIEW

Burnout is a gradual breakdown resulting from sustained, unmanaged stress, first identified in service-oriented professions like healthcare [2]. The World Health Organization now classifies burnout as an occupational phenomenon [5], and its relevance in high-intensity fields like software development is increasingly recognized. In fast-paced software environments, especially in startups, developers often face high cognitive demands, ambiguous roles, and long working hours—all of which are strongly associated with burnout symptoms [1].

Previous studies have investigated burnout primarily using surveys and interviews [3], which are retrospective, subjective, and difficult to scale. While valuable, these methods do not allow for real-time or early-stage detection. Recent work has explored empirical proxies for developer well-being. For example, sentiment analysis has been applied to commit messages to gauge morale [13], and social coding dynamics on platforms like GitHub have been studied to understand collaborative stressors [6]. However, these studies often treat developers as a homogenous group, and few integrate time-based behavioral features such as overwork patterns, contribution streaks, or inactivity gaps.

Methodologically, prior research overlooks several behavioral signals relevant to burnout [6] found that commit streaks longer than 10 consecutive days are highly anomalous in open-source repositories, suggesting that persistent activity without rest may signal overexertion. Similarly, WHO and ILO findings [8] show that excessive after-hours work—specifically, spending more than 40% of work hours outside typical schedules—is linked to elevated long-term health risks, including cardiovascular disease. These patterns are clearly observable in developer workflows but remain underutilized in longitudinal burnout modeling.

To address this gap, our study integrates temporal commit activity, sentiment trends, and contributor role metadata into a unified framework. We define burnout behaviorally through extended inactivity periods (≥ 2 weeks), drawing from the Danish PUMA Study, which identified 10–14 day work absences as a predictive threshold for burnout-related outcomes [7]. Our operationalization includes early warning signs like abrupt commit drop-offs ($>70\%$) and excessive off-hours coding ($>40\%$), both grounded in occupational health and empirical software engineering research [6], [8].

While burnout has been previously studied in both corporate and open-source ecosystems [4], [6], there is limited research focused on early-stage AI startups. These environments are uniquely characterized by high cognitive load, limited resources, and public accountability via open-source channels [1], [4], [9]. Developers in such settings are often expected to balance rapid iteration with long-term code maintenance, often without the institutional safety nets available in larger

companies [3], [4]. Unlike traditional volunteer-driven open-source communities, AI startups operate under venture funding pressure and tight delivery timelines. Yet, little is known about how burnout manifests under these hybrid pressures. Our study addresses this critical gap by analyzing developer behavior in two open-source AI startups, offering insights relevant for both academic understanding and organizational intervention.

IV. DATASET

We analyzed developer activity from two open-source AI startups: Pear AI and ActiveLoop. These were selected due to their domain relevance, open-source model, distributed teams, and VC backing. Full code and dataset access are provided via our public repository: <https://github.com/thebnbrkr/419proj>.

Data Collection & Preprocessing

Commit metadata was collected using the GitHub REST API over a 10-month period (May 21, 2024 to March 11, 2025). For each commit, we recorded SHA, timestamp, author ID, commit message, and file change metadata.

We filtered out bot commits using a regex heuristic on usernames and emails (e.g., patterns like “bot”, “CI”, “[bot]”), and manually verified common automated commit signatures. Contributors with fewer than 50 commits were also excluded to ensure sufficient data per developer.

Timestamps were converted from UTC to local developer time zones using `pytz` and GitHub/LinkedIn metadata, allowing accurate analysis of after-hours and weekend work.

Sentiment Analysis

Commit messages were processed using two sentiment models: VADER and TextBlob. Both are optimized for short, informal text like commit logs. We used both tools to cross-validate sentiment polarity, ensuring robustness across neutral-heavy and emotionally subtle messages. Despite producing similar aggregate trends, VADER handles negation and punctuation sensitivity more effectively, while TextBlob offers easier integration for phrase-level analysis.

Burnout Labeling

A potential burnout event was labeled when a developer had ≥ 2 weeks of inactivity following a high-commitment period. This threshold is supported by occupational health research such as the Danish PUMA Study [7], which found that 10–14 day absences are predictive of burnout-related sickness outcomes.

Contributor Annotation

From 33 total contributors, we selected 13 most active developers for analysis. These were annotated with roles (e.g., founder, core maintainer, casual contributor) and assigned to time zones based on public profile metadata. This enabled comparison of burnout risk across responsibilities and regions.

TABLE I
SUMMARY OF DATASET CHARACTERISTICS.

Metric	Value
Total commits	400
Number of repositories analyzed	2 (Pear AI and Actueloop)
Unique contributors	33
Contributors analyzed in-depth	13 (most active)
Commit date range	May 21, 2024 – March 11, 2025
Mean commits per contributor	12.1
Median commits per contributor	3

Summary Statistics

V. METHODOLOGY

This study operationalizes burnout as a behavioral and emotional response to sustained overwork and inadequate recovery, using indicators observable in GitHub activity logs. The selected variables—including late-night commits, regional time zone offsets, and contributor roles—are grounded in occupational health theory, which links irregular schedules, emotional strain, and role overload to burnout risk [1], [3], [7], [8].

RQ1: Can commit activity and sentiment analysis predict periods of developer burnout in startup repositories?

To identify pre-burnout patterns, we constructed a time-series analysis pipeline combining behavioral and emotional signals.

Feature Engineering: Weekly-aggregated features were extracted to represent developer work habits:

- **Late-night work:** Commits made between 12:00 AM and 5:00 AM (adjusted to local time) [11]
- **Weekend work:** Commits made on Saturday or Sunday [12]
- **Daily commit spikes:** Days with more than five commits, indicating possible overload [13]
- **Negative sentiment:** Commit messages with VADER sentiment scores below -0.3 [14]
- **Commit variability:** Standard deviation of daily commits per week, reflecting work instability [15]

Burnout Risk Score: We designed a composite score, `burnout_risk_score`, to quantify burnout risk across time:

$$\text{burnout_risk_score} = (LN \times 3) + (WE \times 2) + (DC > 5 \times 2) + (S < -0.3 \times 1.5) + (CV > \text{median}(CV) \times 1)$$

where LN = late-night commit ratio, WE = weekend commit ratio, DC = daily commits, S = sentiment score, and CV = commit variability.

This scoring approach was selected instead of supervised learning due to the lack of ground truth labels for burnout and the goal of enabling continuous monitoring. Modeling burnout as a spectrum rather than a binary classification aligns with occupational health research [2], [7], [12].

Burnout Identification: Burnout episodes were operationalized as ≥ 14 -day inactivity periods following weeks

of intense work activity. This threshold is consistent with behavioral withdrawal models of burnout [7].

Visualization: We generated time-series plots showing:

- Evolution of burnout risk scores
- Sentiment trends in commit messages
- Annotated periods of likely burnout

RQ2: How does a developer's level of responsibility within an open-source AI startup (e.g., founder, core maintainer, occasional contributor) shape their vulnerability to burnout, as reflected in behavioral signals?

To evaluate the relationship between contributor roles and burnout indicators, we categorized developers as founders, core maintainers, or casual contributors.

We compared the following metrics across groups:

- Late-night commit percentage
- Weekend work ratio
- After-hours work share
- Negative sentiment ratio
- Average burnout risk score

Statistical Testing: Due to non-normal distributions and small sample sizes, we used the Mann–Whitney U test to compare role groups. Pairwise tests were conducted for:

- Founders vs. casual contributors
- Core maintainers vs. casual contributors
- Founders vs. core maintainers

This method aligns with prior burnout studies involving small and skewed groups [4], [13].

Qualitative Role Trends: We also examined commit streaks and sentiment patterns qualitatively to identify emotional cues (e.g., "sorry for delay," "quick fix," "need to ship") reflecting stress or urgency [9], [14].

RQ3: How do irregular contribution rhythms—such as late-night or weekend work—correlate with behavioral signals of burnout in open-source AI startups with globally distributed teams?

This question investigates whether asynchronous coordination across time zones influences burnout risk. We framed remote work in terms of coordination stress and time-shifted labor.

Definitions:

- **After-hours work:** Activity outside of 9 AM–5 PM local time
- **Late-night work:** Commits between 10 PM and 7 AM
- **Weekend work:** Saturday and Sunday commits

These features help isolate off-schedule behaviors, which are linked to psychological strain in globally distributed teams [16]–[18].

Regional Assignment: Developers were grouped into Pacific, New York, Eastern, and India clusters based on their GitHub metadata and local time zone, determined using the `pytz` Python library. The regions were Pacific (Regions under the PST Time zone in both North and South Americas), New York Time (Regions under the EDT Time zone in both North

Fig. 3. The average commit message length measured to determine the patterns for RQ1

Unique commit authors per day



Fig. 4. Measuring the unique commit activity for determining RQ1

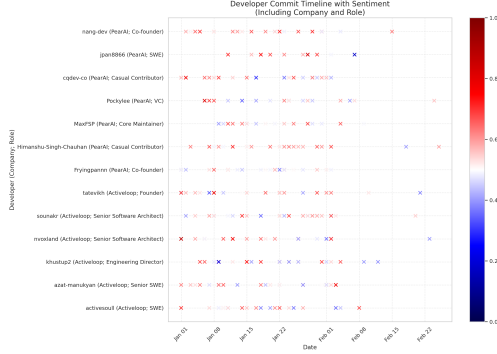


Fig. 5. Developer commit timelines with sentiment color mapping. Each dot represents a commit, colored by sentiment score: red for negative, blue for positive, and white for neutral. Gaps in activity and clusters of negative sentiment indicate potential burnout risk. Developers are sorted top-to-bottom by total commit volume

consistently preceded prolonged inactivity. These findings suggest that while emotional tone may reflect underlying stress in isolated cases, burnout is better predicted by patterns of overwork and poor recovery—particularly those concentrated during non-standard hours.

TABLE III
HIGH BURNOUT RISK DEVELOPERS

Developer	Observations
sounakr	High-volume activity followed by a drop; negative sentiment clusters
Fryingpannn	Bursts of commits with negative tone followed by total inactivity
azat-manukyan	Regular work showing increasingly negative tone before burnout gap
MaxFSP	Red sentiment streaks preceding a complete drop-off in activity

TABLE IV
MODERATE BURNOUT RISK DEVELOPERS

Developer	Observations
khustup2	Sustained work with some negative/neutral sentiment zones
nang-dev	Decrease in commit intensity with mild negative tone
jpan8866	Limited commits with multiple negative messages prior to inactivity

TABLE V
LOW BURNOUT RISK DEVELOPERS

Developer	Observations
nvoxland	Consistent activity with mostly positive sentiment
tatevikh	Minimal emotional volatility; low volume but steady pattern
Pockylee	Infrequent, positive commits; no burnout indicators
Himanshu-Singh-Chauhan	Stable activity; some variance in sentiment but no severe drops
cqdev-co	Balanced commits and sentiment, no intense stress indicators
activesoull	Slight emotional dips but stable output

RQ2: How does a developer’s level of responsibility within an open-source AI startup (e.g., founder, core maintainer, occasional contributor) shape their vulnerability to burnout, as reflected in behavioral signals?

Commit Message Sentiment: All groups’ commit message sentiments were generally neutral on average (average sentiment scores hovering around 0). We observed no statistically significant difference in average sentiment between roles – e.g. founders’ mean sentiment 0.00 vs. casual contributors’ ~ 0.06 ($p \approx 0.45$, Mann–Whitney U). However, sentiment trends hinted at subtle differences. Casual contributors showed a slight tendency toward more positive wording (perhaps because their contributions are often minor improvements or documentation), whereas some core members and founders had mildly negative sentiment streaks. For instance, one casual contributor’s messages became progressively more negative over a short burst (anecdotal), and the project founder’s sentiment averaged slightly negative (-0.003). Overall though, no strong sentiment decline was seen for core or lead developers over time – their commit message tone remained fairly steady. This suggests that textual frustration signals were not markedly different across roles, or were too subtle to distinguish statistically.

Long Commit Streaks and Drop-off Patterns: We found no instances of extreme commit streaks longer than 10 days in a row for any developer. The maximum continuous daily streak observed was 4 days (by a core maintainer), indicating that all contributors took regular breaks at least weekly. This is a positive sign, as relentless daily coding without breaks (which could indicate workaholic behavior or burnout risk) did not occur. We did observe drop-off patterns consistent with differing commitment levels: founders and core maintainers were active during the project’s initial phase and then saw a sharp decline in activity, whereas casual contributors by definition made only brief, infrequent contributions. For example, the project founder committed regularly for a few months and then stopped committing entirely by the project’s later stages, potentially handing off work or experiencing burnout. Core maintainers similarly had concentrated bursts of work (often around feature completion or release deadlines) followed by periods of little to no commits. Casual contributors typically made one-off commits or a short streak of fixes and then dropped off immediately. These patterns align with expect-

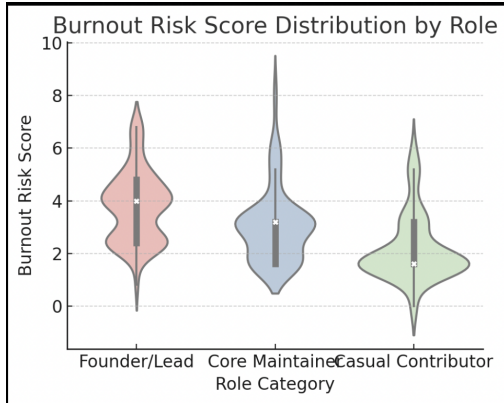


Fig. 6. Distribution of burnout risk scores by role category. Violin plots show that Founder/Lead commits tend to have higher risk scores (median around 4) than Core Maintainers (3) and Casual Contributors (1–2).

tations: founders and core team members carry the project early on and are at risk of burnout-induced withdrawal later, while casual contributors inherently “drop off” after small contributions.

Overall Burnout Risk: We leveraged the `burnout_risk_score` (a composite metric provided per commit) to compare overall risk levels. **Founders/Leads** showed the highest burnout risk scores on average, followed by **core maintainers**, with **casual contributors** showing the lowest.

The distribution of risk scores by role (see Figure 6) illustrates this trend: founder/lead commits are skewed toward higher risk values (median ~ 4.0), compared to core maintainers (median ~ 3.2) and casual contributors (median ~ 1.6).

Statistical tests confirm these differences: the founder/lead group’s risk scores were significantly higher than those of casual contributors ($p \approx 3.7 \times 10^{-14}$) and also higher than core maintainers ($p \approx 2.4 \times 10^{-6}$). Core maintainers likewise had significantly higher risk scores than casuls ($p \approx 9.3 \times 10^{-6}$).

This aligns with the behavioral metrics: heavy after-hours activity correlates with elevated risk. Casual contributors, with minimal sustained workload, showed the lowest burnout risk. Notably, the high proportion of late-night work for founders coincided with higher anomaly scores — supporting the idea that irregular work patterns contribute to burnout risk.

In summary, Founders/Technical leads are at the greatest risk (juggling the most after-hours work and pressure), core maintainers at moderate risk, and casual contributors at comparatively low risk for burnout.

TABLE VI
BURNOUT-RELATED METRICS BY ROLE GROUP.

Metric	Founder/Lead	Core Maintainer	Casual Contributor
Late-night commits (%)	26.4	8.8	1.6
Weekend commits (%)	38.7	46.0	82.7
After-hours commits (%)	36.0	26.9	6.7
Negative messages (%)	6.4	5.8	10.7
Avg. burnout risk score	3.8	3.0	2.2

RQ3: How do irregular contribution rhythms—such as late-night or weekend work—correlate with behavioral signals of burnout in open-source AI startups with globally distributed teams?

We analyzed individual developers across different regions/timezones to examine how remote, asynchronous work patterns relate to burnout indicators. The team was distributed across Pacific (San Francisco/Boulder, UTC−7/ −6), Eastern (Bolivia/Florida/Montreal, roughly UTC−4), and India (Bangalore, UTC+5:30) regions. We converted all commit timestamps to each developer’s local time to assess work hour patterns. Figure ?? shows the local hourly commit distribution for each region.

Working Hours vs. Off-hours: Developers in the Eastern region generally adhered more strongly to a normal work schedule. Over 38% of Eastern region commits occurred during local work hours (9 am–5 pm weekdays), and almost none on weekends (only $\sim 2.6\%$). Their commit activity peaked around late morning to afternoon local time, and virtually no code was committed past midnight local time. This suggests Eastern developers maintained a healthier work-life separation, likely because they were in similar time zones and could collaborate during normal hours.

In contrast, Pacific timezone developers showed more irregular schedules. Notably, Fryingpannn (full-time, Pacific) demonstrated the highest late-night work percentage (13%) among all developers, with only $\sim 17\%$ of their commits within the 9–5 window, while 42% were late-night (post-midnight).

The India-based developer (azat-manukyan) followed a decidedly asynchronous pattern. Only 20% of their commits fell within India’s normal daytime hours. Instead, a large portion were made during Indian late nights and weekends. In fact, 46.7% of this developer’s commits took place on weekends, and 26.7% were late at night (often after midnight local). In Figure ?? (India panel), we see two clusters: one around 12–1 AM (corresponding to early evening in the Americas) and another smaller cluster in the late morning or early afternoon. This indicates the India developer adjusted their schedule significantly—likely to sync with the U.S. team’s working hours (coding late at night local time) and/or to utilize weekend time for contributions.

Founding/Core Members vs. Open-source Contributors: Our analysis revealed distinct patterns between founding/core team members (who work full-time) and open-source contributors. Weekend work was extremely high among founding/core team members across all regions, suggesting that full-time remote status appears to blur work-life boundaries regardless of location. In contrast, many open-source contributors (e.g., Pockylee, tatevikh) tended to contribute either during off-work hours or on weekends, showing significantly lower burnout risk scores (1.6) with minimal off-hours activity. This suggests our burnout risk metrics may not fully capture open-source contribution patterns, as these developers often contribute outside their regular employment hours.

Burnout Risk by Role and Region: The differences in

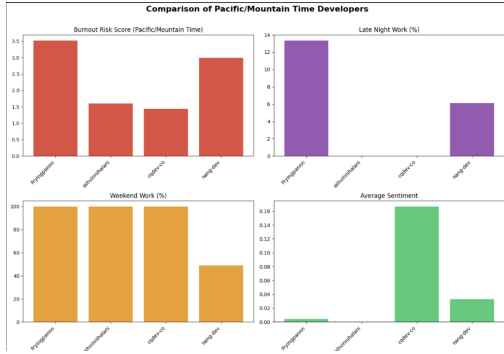


Fig. 7. Commit time distribution (by local hour) for Pacific in Pear AI. The founders (Fryingpannn and nandev are more likely to experience burnout, and work on weekends. The dev casual contributor mostly works during the weekends (ashvinihalini), and cq-dev (employee) is at a lower risk of burnout event after working on weekends.

work patterns translated into varying burnout risk profiles across developers. The India-based open-source contributor (azat-manukyan) showed the highest burnout risk indicators (average risk score 4.1), primarily due to frequent late-night (26.7%) and weekend (46.7%) work patterns. Founding/core team members (nang-dev, Fryingpannn, cqdev-co, jpan8866) consistently showed moderate to high burnout risk, while other open-source casual contributors demonstrated significantly lower risk.

Core maintainers and founding team members (e.g., Fryingpannn, MaxFSP) showed consistently higher burnout risk regardless of region, suggesting that role intensity is a stronger predictor than location. The heatmap in Figure ?? combines key risk metrics: Pacific and India regions are “hotter” on at least one dimension. The Pacific region had the highest late-night commit percentage (42.9%), reflecting a heavy after-hours workload. The India-based open-source contributor had by far the highest weekend work (46.7%) and also substantial late-night work (26.7%), despite not being a core team member. The Eastern group, with their more regular schedules, had minimal weekend coding and relatively lower late-night work (28.2%, chiefly from one founder working late).

Interestingly, full-time Pacific developers showed similar risk levels to full-time Eastern developers, suggesting that full-time status and core maintainer roles may be more determinative of burnout risk than geographic location alone. However, remote coordination appears to impact risk more than remote work alone—developers adjusting schedules for asynchronous work (like the India developer) show higher stress signals.

Significant Patterns: The differences between contribution patterns were substantial. For example, the India-based open-source contributor had a much higher weekend commit percentage than the Eastern group (46.7% vs. 2.6%)—a notable contrast. While our sample sizes are small, the consistency of patterns across multiple metrics adds confidence: time-shifted work and core maintainer status correlate with higher burnout signals, though we recognize that these metrics may not fully capture the intentional nature of off-hours contributions from

open-source developers.

Workload Imbalance and Geographic Isolation: Our analysis suggests that workload imbalance due to geographic isolation (e.g., India, Eastern) can lead to elevated burnout risk indicators even if overall commit volume is lower. This is particularly evident in the case of the India-based open-source contributor, who despite potentially having fewer total commits than core maintainers in Pacific or Eastern regions, still exhibited patterns that would typically indicate high burnout risk.

These findings suggest that contributors working in highly asynchronous conditions (e.g., a contributor from Bangalore with a 10–13 hour time difference) may need to adapt their schedules in ways that register as burnout risk under our metrics. However, for open-source contributors, these patterns might represent deliberate choices rather than imposed demands. In contrast, developers who could collaborate during normal hours and disconnect on weekends showed lower risk indicators.

In summary, the data suggests that distributed, asynchronous work can register higher burnout risk indicators, especially for contributors far removed in time zone who adjust by working nights and weekends. Core maintainer and founding team member status appears more predictive of burnout risk than geographic location alone, but geographic isolation can exacerbate these patterns. For open-source contributors, their contribution patterns may naturally fall outside typical working hours, which our metrics might misinterpret as burnout risk. Teams should be mindful of workload allocation for remote members—encouraging regular hours and mitigating the need for frequent late-night or weekend work may help reduce burnout risk in distributed teams, particularly for core members.

VII. DISCUSSION

This study investigates the feasibility of detecting developer burnout in the demanding environment of remote, open-source AI startups by analyzing behavioral signals from GitHub activity. The three research questions explored burnout predictability, the role of responsibility, and the impact of distributed, asynchronous work patterns. The findings point to actionable behavioral markers and practical strategies for early burnout detection and mitigation.

RQ1: Can Commit Activity and Sentiment Predict Burnout?

Traditional self-report methods often fail to capture burnout in real-time. This research explored whether objective behavioral signals—specifically commit activity and sentiment—could serve as predictors of burnout. The results indicate that while commit message sentiment alone was inconsistent, certain behavioral patterns were highly indicative of burnout.

A key discovery was the Surge-and-Stop Pattern (SSP): short bursts of intense commit activity, frequently during non-standard hours, followed by extended inactivity periods (≥ 2 weeks). This aligns with occupational stress theory, which

views burnout as a process involving overexertion followed by withdrawal. Additional indicators such as reduced commit volume after peaks, shortened commit messages, and a concentration of work on a few individuals also correlated with higher burnout risk scores.

These findings suggest that behavioral data from version control systems can offer a scalable, low-intrusion complement—or alternative—to survey-based assessments. Tools that monitor for high-risk patterns (e.g., SSP, persistent off-hours work, abrupt inactivity) could enable early, automated alerts for burnout risk, supporting timely interventions. Importantly, the focus should shift from sentiment alone to the rhythms and intensity of work behavior.

RQ2: How Do Developer Roles Affect Burnout Risk?

Drawing on organizational behavior theory, this question examined how different levels of responsibility—founder/lead, core maintainer, and casual contributor—shape vulnerability to burnout. The analysis revealed significantly higher burnout risk scores for founders and core maintainers compared to casual contributors. These roles showed more frequent late-night and weekend commits, as well as a higher proportion of total project activity—suggesting sustained workload and pressure.

These behavioral patterns support the hypothesis that leadership roles in high-pressure environments come with substantial psychological costs. Although sentiment did not significantly differ across roles, the behavioral data highlighted the intensity and uneven distribution of responsibility in early-stage AI startups.

Startups must recognize that individuals in high-responsibility roles face disproportionate burnout risk. Lean structures and the absence of formal support systems exacerbate this issue. Practical strategies should focus on clear role definitions, balanced workload distribution, and creating a culture of psychological safety. By proactively supporting those in central roles, startups can enhance both team well-being and long-term sustainability.

RQ3: What Is the Impact of Irregular Rhythms and Distributed Teams?

Building on behavioral science linking irregular routines and insufficient recovery to mental health issues, this question explored how asynchronous work rhythms affect burnout in distributed teams. The findings show that the core challenge lies not in remote work itself, but in the temporal misalignment caused by significant time zone differences.

Developers working across large time gaps (e.g., India-US) displayed the highest proportions of late-night and weekend commits relative to their local time—likely attempts to synchronize with teammates. These contributors also showed the strongest burnout risk signals. In contrast, developers working in more temporally aligned teams maintained more regular work patterns and exhibited lower risk.

For distributed teams, particularly in startups, managing asynchronous collaboration is essential. Mitigation strategies include minimizing expectations for real-time overlap,

establishing clear communication protocols, setting healthy response-time norms, and designing workflows that reduce cross-timezone dependencies. Additionally, care must be taken when interpreting off-hours contributions, especially in open-source contexts, where such patterns might reflect personal preference rather than stress.

Synthesis and Implications

Taken together, these findings suggest that burnout in globally distributed AI startups is primarily driven by behavioral intensity and misaligned rhythms, rather than subjective sentiment or remote work itself. Observable patterns—such as the Surge-and-Stop Pattern, persistent off-hours activity, and high concentration of responsibility—offer a practical lens for early burnout detection.

By leveraging passively collected behavioral signals, teams can move beyond reactive surveys and toward proactive, data-informed well-being practices. This approach can help identify at-risk individuals early and inform targeted interventions, contributing to healthier, more resilient development cultures in the high-pressure landscape of AI-driven software innovation.

VIII. THREATS TO VALIDITY

As with any empirical study, our findings are subject to several validity threats that may affect their interpretation, reliability, and generalizability. We organize these threats into four standard dimensions: construct validity, internal validity, external validity, and conclusion validity.

Construct Validity

This study infers burnout from behavioral patterns rather than self-reports, raising questions about whether our measurements accurately reflect the construct of “burnout.”

- *Operational Definition:* Burnout was operationalized as a 14-day or longer inactivity period following an intense activity surge, based on prior occupational health literature. While grounded in theory, this threshold may not capture more subtle forms of disengagement, such as reduced productivity or emotional exhaustion that do not lead to complete inactivity.

- *Burnout Risk Score:* Our composite `burnout_risk_score` combines multiple heuristics (e.g., late-night work, message sentiment, activity drops). The weights and thresholds used, while literature-informed, are not empirically validated and may not generalize across teams or domains.

- *Sentiment Analysis:* Commit messages are typically brief and technical, limiting the granularity and reliability of sentiment interpretation. Although we used two tools (TextBlob and VADER) for robustness, they may fail to capture nuanced emotional states, particularly for neutral or terse messages.

- *Role and Time Zone Inference:* Developer roles and local time zones were inferred from repository metadata and contribution patterns, which may introduce inaccuracies—especially for geographically mobile contributors or those using pseudonyms.

Internal Validity

While associations between behavior and burnout signals were observed, causality cannot be firmly established.

- *Correlation vs. Causation:* The study identifies patterns (e.g., surge-and-stop, late-night commits) that correlate with inactivity, but cannot determine whether these behaviors cause burnout or are symptoms of other confounding factors (e.g., shifting responsibilities, project deadlines).

- *Uncontrolled Confounders:* Personal factors such as mental health history, job satisfaction, external stressors, or life events were not available and could influence commit behavior independently of burnout.

- *Selection Bias:* We limited our analysis to developers with ≥ 50 commits and manually filtered out bots. This may bias the sample toward more active contributors or misclassify edge cases (e.g., interns, part-time contributors).

External Validity

The narrow scope of the study imposes constraints on generalizability.

- *Sample Scope:* The dataset includes contributors from only two early-stage, open-source AI startups. Cultural norms, work practices, and tool usage in these contexts may differ significantly from larger corporations, non-AI domains, or non-open-source teams.

- *Visibility Bias:* Behavior in public repositories may be influenced by the visibility of work or community engagement norms, which might not translate to private or enterprise contexts.

- *Temporal Limitations:* The data spans 10 months, which may not capture the full evolution of burnout cycles, especially in long-term projects or seasonal development phases.

Conclusion Validity

While the results offer meaningful trends, several factors limit statistical robustness.

- *Sample Size and Statistical Power:* The in-depth analysis covers only 13 developers. This limits power for subgroup analysis and increases sensitivity to outliers, even though non-parametric tests were used.

- *Data Quality:* Inaccuracies in time zone inference, role annotation, or sentiment classification can introduce noise, potentially weakening or distorting statistical relationships.

While we took steps to mitigate these threats—such as using multiple sentiment tools, grounding heuristics in theory, and applying appropriate statistical tests—there remains a need for validation through larger, more diverse samples and mixed-methods approaches. Future work could incorporate self-reported burnout data, interviews, and longitudinal tracking to triangulate and strengthen the behavioral signals identified in this study.

IX. CONCLUSION AND FUTURE WORK

Conclusion

This study explored manifestations of developer burnout in the unique context of early-stage, remote AI startups using

a Mining Software Repositories (MSR) approach. Motivated by the need for objective, early indicators beyond traditional surveys, we analyzed behavioral signals—commit activity patterns, sentiment shifts, contribution rhythms, and work timing—from the open-source repositories of two such startups.

Our findings highlight that burnout signals in these high-pressure, innovation-driven environments are multifaceted. We demonstrated the feasibility of using repository-based behavioral data to detect potential burnout risks, identifying patterns such as "surge-and-stop" activity and persistent off-hours work as stronger indicators than sentiment alone. Notably, burnout vulnerability was strongly associated with developers' roles and levels of integration within the project, echoing the concept of role overload. Founders and core contributors exhibited higher risk markers, reflecting the intense pressure and responsibility they bear in resource-constrained environments. In contrast, open-source contributors, despite their irregular hours, showed lower risk signals—suggesting a potential buffering effect of autonomy and voluntary participation. Furthermore, our analysis of distributed work patterns indicates that the strain may stem less from remote work itself and more from asynchronous collaboration across significant time zone differences, which correlated with higher risk signals.

This research underscores the value of passive, behavioral data for understanding developer well-being. By focusing on AI startups, we examined a setting where speed, innovation, and limited resources converge to intensify burnout risk. While scalable, our behavioral proxies offer only a partial view of the psychological complexity of burnout and must be complemented by more holistic, validated approaches.

Future Work

This exploratory study opens several avenues for future research, which we outline below:

- *Expand Dataset Scope and Diversity:* To improve external validity, future studies should analyze a broader and more diverse range of organizations—including more AI startups, companies across different domains, and teams with varied collaboration models (open vs. closed source, centralized vs. distributed). Such expansion would support comparative analyses to distinguish universal burnout indicators from context-specific patterns and improve statistical robustness.
- *Refine and Validate Burnout Metrics:* The composite burnout risk score and its behavioral inputs require refinement. Future work should explore advanced linguistic features, incorporate signals from other developer interactions (e.g., code reviews, issue comments, communication platforms), and consider physiological or wearable data where ethically feasible. Crucially, these behavioral proxies must be validated against established psychological measures (e.g., the Maslach Burnout Inventory) and qualitative inputs such as interviews or self-reports. Techniques like cross-validation and bootstrapping should also be employed to improve model reliability.

- *Design and Evaluate Ethical Intervention Tools*: The long-term goal is to support proactive, ethical burnout mitigation. Future research should focus on building prototype tools (e.g., dashboards, IDE plugins) that provide developers and teams with actionable insights. However, ethical considerations—especially around privacy, transparency, and the potential for misuse—must remain central. These tools should prioritize support over surveillance, helping teams foster healthier development practices without punitive implications.

By addressing these directions, future research can build on our findings to develop more robust, validated, and ethically responsible approaches for monitoring and improving developer well-being in high-intensity software environments.

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